

Naghmeh (Melody) Nazar

AFFECTIVE COMPUTING | HEALTH AI | HUMAN-CENTERED MACHINE LEARNING

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

RESEARCH PROFILE

Machine-learning researcher with an M.S. in Computer Engineering from California State University, Northridge, studying speech emotion recognition. Work combines audio signal processing and ResNet-inspired neural networks. Interested in robust multimodal sensing, affective computing, and responsible health AI, with a focus on domain shift, subgroup performance, and reproducibility.

EDUCATION

M.S. in Computer Engineering, California State University, Northridge

Aug 2024 - Aug 2026

- GPA: 3.44/4.0
- Thesis: Real-Time Speech Emotion Recognition Using Optimized Deep Residual Networks (ResNet)
- Advisor: Prof. Shahnam Mirzaei

B.S. in Computer Engineering, University of Science and Technology of Mazandaran

Aug 2016 - Mar 2021

MANUSCRIPT & PRESENTATION

- Research manuscript based on the M.S. thesis in speech emotion recognition.
- Presented the speech emotion recognition research at the 40th Annual CSU Student Research Competition.

RESEARCH EXPERIENCE

Real-Time Speech Emotion Recognition - M.S. Thesis, CSUN

2024 - 2026

- Developed a 2D ResNet-inspired CNN prototype with neutral, happy, sad, and angry output labels.
- Processed RAVDESS and TESS speech at 16 kHz and generated 128 x 128 log-Mel spectrograms. The current label mapping groups RAVDESS calm recordings with neutral.
- Research focuses on speech emotion classification and reproducible preprocessing. Speaker-independent and separate held-out evaluation remain to be established.
- Implemented model training with Adam optimization, dropout, and residual connections. Further work includes controlled data partitioning and repeatable evaluation.
- Investigating practical inference for speech-based applications; latency and real-world generalization are not yet established. No clinical validation is claimed.

Persian COVID-19 Social Behavior Analysis - NLP Research

2021

- Analyzed Persian-language social-media posts using sentiment classification, topic modeling, and temporal trend analysis.

SELECTED PROJECTS

CCTV Product-Support Chatbot

Python, Dialogflow, Google Cloud

- Designed an intent-driven assistant that guides users through CCTV product selection and support questions.
- Integrated Dialogflow NLP with cloud-backed services and structured conversational flows for real-time interaction.

GenAI Email & Scheduling Agent

n8n, Gmail, Airtable, Google Calendar, structured JSON

- Built an n8n prototype connecting email classification, business-context retrieval, and response generation. Integrated Gmail, Airtable, OpenAI, and Google Calendar nodes in the exported workflow.
- Used structured outputs and API integrations to connect email and scheduling actions. Exported the workflow to document its node configuration and connections.

PROFESSIONAL EXPERIENCE

Senior Frontend Developer, Golrang Industrial Group - Tehran, Iran

Jul 2022 - Jul 2024

- Developed and maintained production web and mobile applications using React, React Native, Vue.js, TypeScript, and modern state-management patterns.
- Improved responsiveness and maintainability through profiling, caching, code splitting, debugging, and systematic code review.

React Developer, Erole.ir - Tehran, Iran

Jun 2021 - Jun 2022

- Built responsive React/Redux applications and integrated REST APIs and SignalR-based real-time features.

TECHNICAL SKILLS

- **Machine Learning:** Python, TensorFlow, Keras, scikit-learn, NumPy, Pandas, CNNs, ResNet, GRU/RNN, classification, evaluation
- **Signal & Behavioral Data:** audio preprocessing, Mel-spectrograms, MFCCs, time-frequency analysis, affective computing, NLP, topic modeling
- **Research Tools:** SciPy, Librosa, Jupyter, Google Colab, Git, Linux, reproducible experiment workflows
- **Engineering:** C++, JavaScript, TypeScript, React, Vue.js, Node.js, REST APIs